

Part II

Functions of Several Variables.

- Domains in $\mathbb{R}^2$ and $\mathbb{R}^3$
- Continuous functions on domains in $\mathbb{R}^2$ and $\mathbb{R}^3$
- Partial derivatives.
- Differentiability. Chain Rule
- Jacobians
- Taylor's Thm and Maxima Minima of functions of several variables
- Lagrange Multipliers

Chapter V

Open sets and closed sets in $\mathbb{R}^2$ and $\mathbb{R}^3$.

points in $\mathbb{R}^2$ or $\mathbb{R}^3$ will be denoted by letters x, y or v ; x, y or p, q etc; so let $p \in \mathbb{R}^2$ (or $\mathbb{R}^3$)

and $r > 0$. (which means r is a real number!) $>$ and $<$ are defined only for real numbers)

disc

$B_r(p)$ = Open ball of radius r centred at p (in $\mathbb{R}^2$)

$$B_r(p) = \{ (x_1, x_2) / (x_1 - p_1)^2 + (x_2 - p_2)^2 < r^2 \}$$

$$B_r(p) = \{ (x_1, x_2, x_3) / (x_1 - p_1)^2 + (x_2 - p_2)^2 + (x_3 - p_3)^2 < r^2 \}$$

For $x = (x_1, x_2)$ we denote $p = (p_1, p_2)$

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The distance between p and x by $\|x - p\|$ (similar notation in $\mathbb{R}^3$)

So $B_r(p)$ is actually a disc while in $\mathbb{R}^2$

$\mathbb{R}^3$ it is a ball.

However we shall call it a ball in both $\mathbb{R}^2$ and $\mathbb{R}^3$. ==

Boundary of $B_r(p)$ is the sphere

$$\|x - p\| = r \quad \text{or}$$

$$\left\{ (x_1, x_2, x_3) / (x_1 - p_1)^2 + (x_2 - p_2)^2 + (x_3 - p_3)^2 = r^2 \right\}$$

Boundary of $B_r(p)$ will be denoted by $\partial B_r(p)$

This is the Standard notation

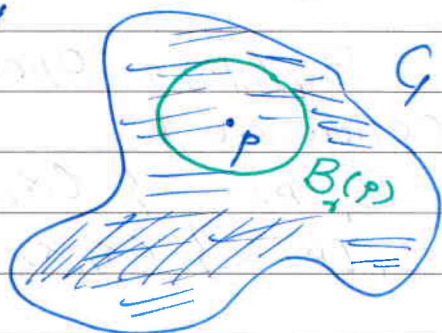
Open sets in $\mathbb{R}^2$ or $\mathbb{R}^3$.

Let G be a subset of $\mathbb{R}^3$ (For both $\mathbb{R}^2$ and $\mathbb{R}^3$ the def. is similar)

G is said to be an open set if for each $p \in G$, there is a $r > 0$ such that $B_r(p) \subset G$

(See the figure)

ϕ is by convention an open set



Examples: $\mathbb{R}^2$ itself is an open set in $\mathbb{R}^2$

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The first quadrant

$Q^+ = \{ (x_1, x_2) / x_1 > 0, x_2 > 0 \}$ is an open set in $\mathbb{R}^2$

The positive Orthant-

$\{ (x_1, x_2, x_3) / x_1 > 0, x_2 > 0, x_3 > 0 \}$ is an open set in $\mathbb{R}^3$.

Ex: prove that $\{ (x_1, x_2, x_3) / x_1 + x_2 + x_3 > 0 \}$ is open in $\mathbb{R}^3$ (this is an open half space)

An open ball is an open set.

Let $B_r(p)$ be an open ball and

take any $q \in B_r(p)$

$$\|p - q\| < r$$

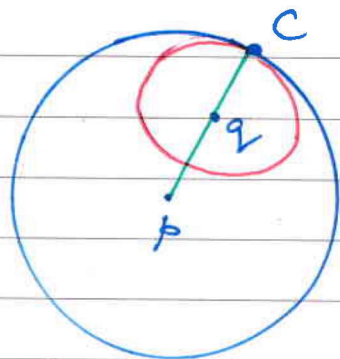
Let

$$s = r - \|p - q\|$$

Draw a ball of radius s centred at q

(depicted in red). This red ball touches the boundary $\partial B_r(p)$ internally

So the red ball $B_s(q) \subset B_r(p)$



Note that the point of tangency C

is not in $B_s(q)$ nor in $B_r(p)$

We are talking about OPEN Ball

It would be EXTREMELY unpleasant to check that

$$\left\{ (x_1, x_2) / \frac{x_1^2}{4} + \frac{x_2^2}{9} < 1 \right\}$$

is an open set - though it ought to be intuitively clear.

Even more unpleasant would be to figure out whether

$$\left\{ (x_1, x_2) / \frac{x_1^3}{73} - \frac{x_2^4}{37} < 1 \right\}$$

is open in $\mathbb{R}^2$

We shall see how to get around these difficulties with ease.

A sequence (v_n) in $\mathbb{R}^2$ (or $\mathbb{R}^3$) is said to converge to $v \in \mathbb{R}^2$ (resp $\mathbb{R}^3$) if $\|v_n - v\| \rightarrow 0$ as $n \rightarrow \infty$.

The points v_n come arbitrarily close to v .

Exercise . Prove that

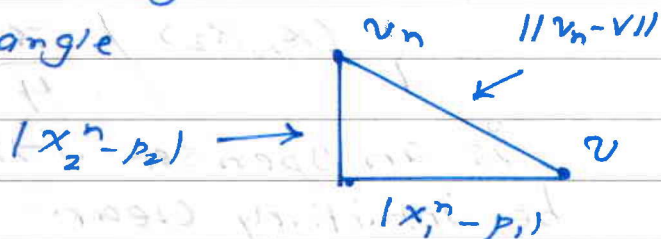
Let $v_n = (x_1^n, x_2^n)$

$v_n \rightarrow v = (p_1, p_2)$ iff the individual real sequences

(x_1^n) and (x_2^n) converge to p_1, p_2 respectively

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This follows at once using Sandwich
Thm and the familiar result that the
hypotenuse is the longest side in a
right-angle triangle



Def: A set F in $\mathbb{R}^2$ is closed
if $\mathbb{R}^2 - F$ is open

(Same def in $\mathbb{R}^3$)

Thus $\emptyset$ open $\Rightarrow \mathbb{R}^2$ closed

$\mathbb{R}^2$ open $\Rightarrow \emptyset$ is closed

So $\mathbb{R}^2, \emptyset$ are both open and closed.

Ex: Show that

$G = \{ (x_1, x_2) / x_1^2 + x_2^2 > 1 \}$
is open in $\mathbb{R}^2$

Show that the Union of two open sets
is open (What about union of 3 open
sets? Union of infinite collection of open
sets ??)

In particular $G \cup B_1(0)$
is open in $\mathbb{R}^2$

So its complement
 $\{ (x_1, x_2) / x_1^2 + x_2^2 = 1 \}$ is
Closed in $\mathbb{R}^2$

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The closed ball $\bar{B}_r(p)$
 $= \{ (x_1, x_2) / (x_1 - p_1)^2 + (x_2 - p_2)^2 \leq r^2 \}$
is closed. Since the complement

$\{ (x_1, x_2) / (x_1 - p_1)^2 + (x_2 - p_2)^2 > r^2 \}$ is
an open set (How?)

Thm 5.1 : Let F be a nonempty
subset of $\mathbb{R}^2$ (case with $\mathbb{R}^3$ similar)

F is closed if and only if the

following condition holds:

Whenever a seq (v_n) of points $v_n \in F$
converges to v , the limit v also lies
in F .

Let us use this thm right away.

$$G = \{ (x_1, x_2) / \frac{x_1^2}{4} + \frac{x_2^2}{9} < 1 \}.$$

prove G is open. Need to prove $\mathbb{R}^2 - G$ is
closed. Now $\mathbb{R}^2 - G = \{ (x_1, x_2) / \frac{x_1^2}{4} + \frac{x_2^2}{9} \geq 1 \}$

Take a sequence (v_n) in $\mathbb{R}^2 - G$

converging to $v = (p_1, p_2)$

$$v_n = (x_1^n, x_2^n) \in \mathbb{R}^2 - G$$

$$\therefore \frac{(x_1^n)^2}{4} + \frac{(x_2^n)^2}{9} \geq 1$$

Taking limit as $n \rightarrow \infty$

$$\frac{p_1^2}{4} + \frac{p_2^2}{9} \geq 1 \quad \therefore v = (p_1, p_2) \in \mathbb{R}^2 - G$$

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Thus, $\mathbb{R}^2 - G$ is closed and we conclude
 G is open

Exercise: Prove that

$$G = \left\{ (x_1, x_2) / \frac{x_1^4}{73} - \frac{x_2^{10}}{37} < 1 \right\}$$

is open set

Optional Exercise: Prove Thm 5.1

It is not difficult.

Need to Concentrate on the definitions only.

Use Thm 5.1 to prove that the half space
 $\{ (x_1, x_2, x_3) \in \mathbb{R}^3 / x_1 + x_2 + x_3 > 0 \}$ is
an open set (*)

nonempty

Convex sets: A nonempty subset C of $\mathbb{R}^2$
(or $\mathbb{R}^3$) is said to be

Convex if given $x, y \in C$ the points
 $tx + (1-t)y : 0 \leq t \leq 1$
all lie in C

That is, given $x, y \in C$ the line
segment joining x and y lies entirely in C

Ex: $\mathbb{R}^2$, $\mathbb{Q}^+$ are convex. The half space
(*) is convex (proof?)

Show that $\mathbb{R}^2 - (0,0)$ is not convex

Is the set $\{ (x_1, x_2) / \frac{x_1^2}{4} + \frac{x_2^2}{9} < 1 \}$ convex?
proof??



Rectangles and (Rectangular) Boxes

A closed rectangle $\bar{R}$ in the plane is given by

$$\bar{R} = \{ (x, y) \in \mathbb{R}^2 / a \leq x \leq b; c \leq y \leq d \}$$

or $[a, b] \times [c, d] \quad : (a < b, c < d)$

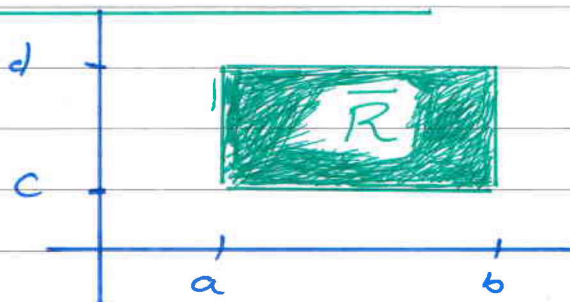
This is a closed bounded rectangle

closed

A box in $\mathbb{R}^3$ is simply

$$\{ (x_1, x_2, x_3) \in \mathbb{R}^3 / a_1 \leq x_1 \leq b_1; a_2 \leq x_2 \leq b_2 \text{ and } a_3 \leq x_3 \leq b_3 \} \quad (\text{Sketch?})$$

(A closed bounded box in $\mathbb{R}^3$)



$$\text{Box: } [a_1, b_1] \times [a_2, b_2] \times [a_3, b_3]$$

Show that a closed bounded rectangle / Box is Convex

Open rectangle / box This is the set-

$$R = \{ (x, y) \in \mathbb{R}^2 / a < x < b; c < y < d \}$$

Define the notion of Open box

Open rectangles / boxes are Convex

Another Standard domain we may use often:



The Standard closed square $\bar{Q}$

$$\bar{Q} = [0, 1] \times [0, 1]$$

Special case of a closed rectangle

Open (standard) sq: $Q = (0, 1) \times (0, 1)$

Open Cube in $\mathbb{R}^3$ $C = (0, 1) \times (0, 1) \times (0, 1)$

Closed Cube in $\mathbb{R}^3$: $\bar{C} = [0, 1] \times [0, 1] \times [0, 1]$

Exercise: Let P, Q, R be three non
collinear points in $\mathbb{R}^2$

define precisely the closed Triangle determined
by P, Q, R

Define the open triangular region
determined by P, Q, R

Are these Convex?

Proof of Thm 5.1: A nonempty set
 $F \subset \mathbb{R}^2$ (or $\mathbb{R}^3$) is

given. The foll. are equivalent

(i) F is closed

(ii) If (v_n) is a sequence of points of F
s.t. $v_n \rightarrow v$ then $v \in F$.

proof: (i) $\Rightarrow$ (ii) Assume (i) namely
 $\mathbb{R}^2 - F$ closed.

Suppose (v_n) is a seq of pts in F s.t. $v_n \rightarrow v$

If $v \notin F$ then $v \in \mathbb{R}^2 - F$ (open)



So - there is a radius $r > 0$ such that -
the open disc $B_r(v) \subset \mathbb{R}^2 - F$

Now $v_n \rightarrow v \Rightarrow$ there exists $n_0 \in \mathbb{N}$

s.t. $v_n \in B_r(v)$ for all $n \geq n_0$

($\text{dist}(v_n, v) < r$ for all $n \geq n_0$).

i.e. $v_n \in \mathbb{R}^2 - F \quad \forall n \geq n_0$

Contradiction since $v_n \in F$.

proof (ii) $\Rightarrow$ (i) Assume (ii)

We have to show $\mathbb{R}^2 - F$ is open

So let $v \in \mathbb{R}^2 - F$. Need to show for

some $r > 0$, the disc $B_r(v) \subset \mathbb{R}^2 - F$

If no such r exists

- then, in particular $r = 1/n$ won't work

i.e. $B_{1/n}(v) \not\subset \mathbb{R}^2 - F$

i.e. $\exists v_n \in B_{1/n}(v)$ s.t. $v_n \in F$

Thus, we have a seq. (v_n) of pts of F

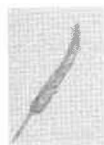
such that $\text{dist}(v_n, v) < 1/n$

$\therefore v_n \rightarrow v$

By (ii) - this means $v \in F$

Contradiction since $v \in \mathbb{R}^2 - F$

Case when $F \subset \mathbb{R}^3$ is similar.



Let $f: I \rightarrow \mathbb{R}$ be Convex

$g: I \rightarrow \mathbb{R}$ be Concave

(I open interval in $\mathbb{R}$)

Assume $g(x) > f(x)$ for all $x \in I$

Then $G = \{ (x, y) \in \mathbb{R}^2 / x \in I, f(x) < y < g(x) \}$

is Convex.

(Tutorial problem)

Show that the region

$\{ (x, y) / x \in I, y > f(x) \}$

is Convex

(Sketch)

$\{ (x, y) / x \in I, y < g(x) \}$ is Convex.

(Sketch)

Intersection of two Convex sets is Convex.

